



1.3 Objective

The objective of this Pre-feasibility study is for Solar resource assessment of identified areas under HIL's manufacturing facilities across three states in India namely, Punjab, Maharashtra, and Kerala.



2 Executive Summary

Project : Pre-feasibility Study for Solar
Resource Assessment for HIL Sites

Project proponent : PTC India Limited

Plant Location : HIL Facilities in Punjab,
Maharashtra, and Kerala.

Geographical location : Locations considered in this study
are:

Location	Est. Solar capacity proposed for installation (in kWp)
Udhyogmandal	603.5
Rasayani	1530
Bhatinda	17.6
Total	2151.1

Solar PV Technology : Monocrystalline Silicon

PV Module Capacity : 345Wp (for Resource Assessment purpose)

PV Module Mounting Structure : Fixed type

Annual degradation (%) : First Year : 2.5%
After First Year : 0.70% subsequent year

Electrical Interconnection : Existing electrical panel at the
individual building/ nearest existing electrical room



3 Approach & Methodology

HIL has identified and shared the details of the facilities including the details of land and building facilities where solar energy could be harnessed by installing solar PV modules. HIL's three facilities namely, Rasayani (Maharashtra), Udhyogmandal (Kerala) and Bhatinda (Punjab) were covered in the study. Identification of prospective sites were done as per the instructions and guidance from HIL.

Based on the "Google Earth", the shadows are assessed and accordingly PV modules are planned for installation on the proposed locations for optimum solar energy generation.

Solar yield estimation and shadow analysis is based on the Helioscope software with Meteronorm data. Typical Helioscope report for each of the identified site for which study is undertaken is attached for each respective location in this pre-feasibility report.

Technology selection was made for the optimum use of the type of PV module to be adopted. Components of the solar PV modules are brought out in the report for better understanding. Various tools and software available for solar resource assessment is mentioned in the Article below.

Most of the proposed sites are undulated open land parcels which needs leveling and removal of shrubs and small trees and plants before taking up the site for solar plant installation. Further, the most of buildings across all three are either very old and having inclined rooftop or surrounded with trees rendering them not suitable for rooftop solar plant installation.

Finally, business models available in the market and best suited model is brought out in the report.

4 Technology Selection

4.1 Introduction

Assessment of solar technology covers mainly Solar PV modules which constitute about 60% of system cost.

4.2 PV Modules

The following solar PV technologies are prevalent in the industry for large scale commercial solar power projects.

Thin Film

Crystalline Silicon Photovoltaic (C-Si)

Appearance of thin film, mono silicon and poly silicon cells are shown below:



Appearance of thin film, mono silicon & poly silicon cells

Solar PV technologies prevalent in the industry are:

- Thin Film
- Crystalline Silicon Photovoltaic (C-Si)
- Mono PERC
- Bifacial

a) Thin Film Technology

Thin-film modules are made by coating and patterning entire sheets of substrate, generally glass or stainless steel, with micron-thin layers of conducting and semiconductor materials, followed by encapsulation. The most common thin-film semiconductor materials are cadmium

telluride (CdTe), amorphous silicon (a-Si), and alloys of copper indium gallium diselenide (CIGS). The semiconductor layer is a few microns (μm) thick, which is about 100 times thinner than current c-Si cells thin film modules.

Amorphous silicon is a special form of crystalline silicon which is deposited as an extremely thin film and requires only about 1% of the silicon used in crystalline systems. Amorphous Silicon (a-Si) solar can be fabricated at a lower deposition temperature hence permits the use of various low cost flexible substrates by easier processing technique.

Thin-film modules have lower DC efficiencies: about 9%–11% for CdTe, 5%–8% for a-Si, and 10%–12% for CIGS.

b) Crystalline Silicon PV

C-Si PV is made up of a number of Crystalline Silicon solar cells. In Crystalline Silicon technology the following types of crystalline silicon PV are being used.

- Mono-crystalline
- Poly-crystalline.

Mono-crystalline cells are manufactured from single crystal silicon ingots. High capacity Mono-crystalline silicon modules produce higher efficiency of over 17%.

Polycrystalline cells are made by melting raw silicon and pouring it into square mould, which is cooled and cut into perfectly square wafers. Polycrystalline processing offers a lower cost per watt for the power produced. Efficiency of poly crystalline module is about 17% for higher ratings.

c) Mono PERC Modules

PERC (Passivated Emitter and Rear Cells) integrates back surface passivation, front surface advanced passivation and anti-light-induced degradation technologies, inside solar panel rest cells made of silicon. A typical 12V panel will house about 36 cells, with each cell contributing to the panels' overall energy generation. Conventional monocrystalline cells have an emitter layer on the front side and a standard back surface field.

PERC cells, on the other hand, employ two different features not found in conventional cells. These include applying a passivation film on the rear surface of the cell and using chemicals to create tiny pockets in the film that absorb more light. Different manufacturers can achieve this end in a variety of ways, but in each instance, the technique involves adding a dielectric passivation layer to the rear of the solar cell.

The cell's front side receives direct sunlight while the rear side soaks up scattered and reflected light. PERC tech is ideal for rooftops, elevated ground, desert or snowy environments or on-water applications.

d) Bifacial Modules

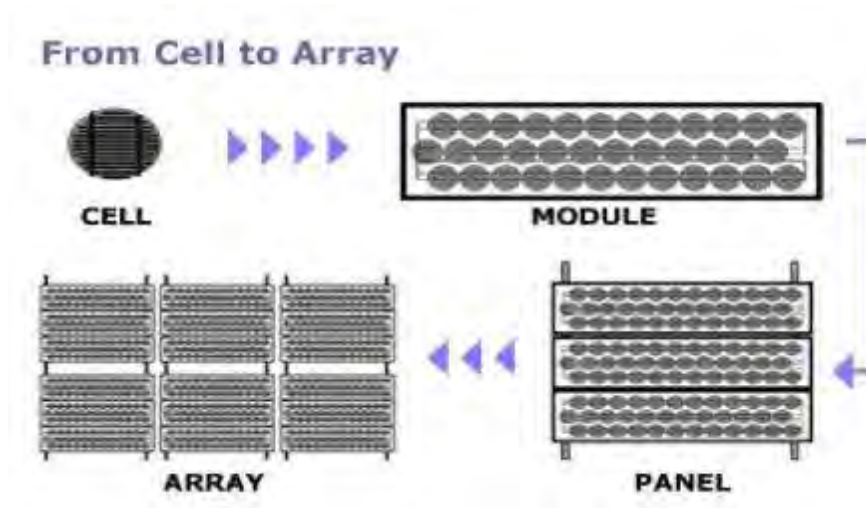
Bifacial modules produce solar power from both sides of the panel. Whereas traditional opaque-back sheeted panels are monofacial, bifacial modules expose both the front and backside of the solar cells. When bifacial modules are installed on a highly reflective surface (like a white TPO roof or on the ground with light –colored stones), some bifacial modules claim up to a 30% increase in production just from the extra power generated from the rear.

Bifacial modules come in many designs; some are framed while others are frameless. Some are dual-glass, and others use clear backsheets.

Most use monocrystalline cells, but there are polycrystalline designs. The one thing that is constant is that power is produced from both sides. There are frameless, dual-glass modules that expose the backside of cells but are not bifacial. True bifacial modules have contacts/busbars on both the front and back sides of their cells.

e) Solar module assembly

The Solar cells are assembled to form a module at manufacturing plants. The finished solar cells are soldered to form ribbons of solar cells, usually in matrix of 12 x 6 configurations. This matrix of solar cells is laminated on to anti-reflective tempered glass, with a plastic back sheet. The glass assembly is then placed in a frame; a junction box is fitted for electrical connection and just forms a module. At site, each of this modules are positioned side by side in series and parallel connections to form array.. The panels are arranged in a row to form an array as depicted below:



f) Comparison of Solar Photovoltaic Technologies

Crystalline Silicon solar cells are widely used in recent years and have higher market share compared to thin film technology. Physical

characteristics of Monocrystalline and Polycrystalline cells are listed below.

No.	Parameter	Polycrystalline – Si	Monocrystalline - Si
1.	Commercially available panel rating range (Wp)	Upto 340	Upto 370
2.	Nominal Efficiency Range (%)	16 - 17.5	17 – 19
3.	Area Requirement per kWp	Around 8 sq.m	Around 6 sq.m
4.	Temperature Coefficient & Losses	Temperature Coefficient is comparatively high. However when the temperature increases beyond certain limit (35 deg.C) the losses incurred are low.	Temperature Coefficient is Low. However when the temperature increases beyond certain limit (35 deg.C) the losses incurred are high.
5.	Irradiance	Works under Normal Radiation. Requires tracking to counter reduction	Works under Normal Radiation. Requires tracking to counter reduction
6.	Cost of Panels	Lower cost per W	Higher cost per W
7.	Power Production under cloud cover or during early / late hours	Higher reduction of power generation or in some cases insignificant generation	Higher reduction of power generation or in some cases insignificant generation
8.	Commercial Availability	Available in adequate levels for commercial installation	Available in adequate levels for commercial installation
9.	Maintenance Requirements	Panels require washing (soap - fortnightly) & rinsing (alternate day) or as per site conditions	Panels require washing (soap - fortnightly) & rinsing (alternate day) or as per site conditions

4.3 Elements of Solar PV panel

A Grid connected Solar Rooftop Photo Voltaic system consists of the following major components are,

- Solar PV modules



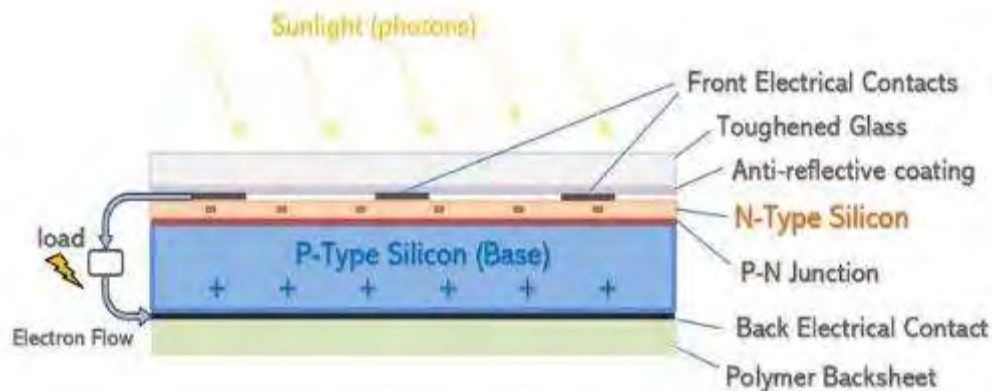
- Module Mounting Structure
- String boxes
- Inverter
- Interconnecting cables
- AC switch board
- Roof space

4.4 Elements of Solar PV

- Silicon PV cells with busbars
- Tempered Glass
- Extruded Aluminum frame
- EVA film
- Backsheet
- Junction box

4.4.1 Silicon PV cells with busbars

Silicon photovoltaic cells or PV cells convert sunlight directly into DC electrical energy. The performance of the solar panel is determined by the cell type and characteristics of the silicon used, with the two main types being monocrystalline and polycrystalline silicon. Additionally the base of the cell can be built using different additives to create either a positive p-type silicon or negative n-type silicon. However there are several different cell configurations available offering different levels of efficiency and performance.



Most residential solar panels contain 60 cells linked together via busbars in series to generate a voltage between 30-40 volts depending on the type of cell used. Larger solar panels used for commercial systems contain up to 72 cells and in turn operate at a higher voltage. The busbars are electrical contacts which interconnect the cells and allow the current to flow in a circuit.

4.4.2 Tempered Glass

The front glass sheet protects the PV cells from the weather and impact from hail or airborne debris. The glass is typically high strength tempered glass which is 3.0 to 4.0mm thick and is designed resist mechanical loads and extreme temperature changes. The IEC minimum standard impact test requires solar panels to withstand an impact of hail stones of 1 inch (25 mm) diameter traveling up to 60 mph (27 m/s). In the event of an accident or severe impact tempered glass is also much safer than standard glass as it shatters into tiny fragments rather than sharp jagged sections.